

Lecture 6: Causal Inference with a Binary Treatment

September 10, 2023

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- ▶ Returning to the CPS example, we will consider 4-year degree holder incomes and income differences between 4-year degree holders and those without a 4-year degree.
- ▶ Even when the overall sample size is fixed (2246 for our example), different random samples will produce different subgroup sample sizes, and this adds a layer of complexity.

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- ▶ Returning to the CPS example, we will consider 4-year degree holder incomes and income differences between 4-year degree holders and those without a 4-year degree.
- ▶ Even when the overall sample size is fixed (2246 for our example), different random samples will produce different subgroup sample sizes, and this adds a layer of complexity.
- ▶ This complexity will not complicate our analysis much but it does complicate the justification for the analysis.

Agenda

- ▶ Introduce formal causal inference with potential outcomes.
- ▶ Demonstrate the identification of average treatment effects with randomized treatment assignment.
- ▶ Discuss constant vs nonconstant treatment effects.

What is formal causal inference?

- ▶ The formal approach reframes the problem of causal inference in terms of counterfactuals which are treated as missing data.
- ▶ This is a natural extension to sampling from populations where the unsampled portion of the population is treated as missing data.
- ▶ This reframing has lead to a revolution in the empirical sciences.

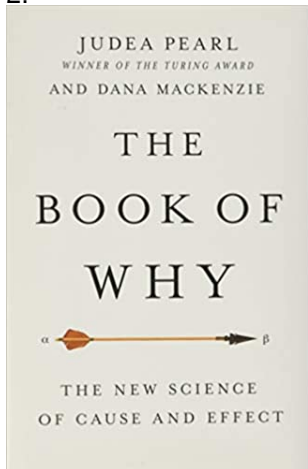
Optional readings for those interested:

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Potential Outcomes and Treatment Effects

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- ▶ D is a treatment variable (e.g. Treatment/Control, college degree/no degree)

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Potential Outcomes and Treatment Effects

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- ▶ D is a treatment variable (e.g. Treatment/Control, college degree/no degree)

Potential Outcomes:

$Y_i(1)$ is the potential outcome for individual i if $D_i = 1$.

$Y_i(0)$ is the potential outcome for individual i if $D_i = 0$.

Treatment Effects:

For example, if patient i will die without the treatment ($Y_i(0) = 0$), but will live with the treatment ($Y_i(1) = 1$), then the treatment effect for individual i is

$$\begin{aligned}\tau_i &= Y_i(1) - Y_i(0) \\ &= 1 - 0 \\ &= 1\end{aligned}$$

Illustrative Example

Suppose we have six adult ADHD patients at a clinic. Two are being treated with stimulants ($D = 1$) and four not ($D = 0$). We are interested in the potential side effects of this treatment on resting heart rate (Y).

i	D_i	Y_i	$Y_i(1)$	$Y_i(0)$	$\tau_i = Y_i(1) - Y_i(0)$
1	1	70	70	?	?
2	1	80	80	?	?
3	0	60	?	60	?
4	0	70	?	70	?
5	0	70	?	70	?
6	0	80	?	80	?

Observed Outcomes vs Potential outcomes

$$Y_i = (1 - D_i) Y_i(0) + D_i Y_i(1)$$

Illustrative Example (red numbers unknown):

i	D_i	Y_i	$Y_i(1)$	$Y_i(0)$	τ_i
1	1	70	70	70	0
2	1	80	80	70	10
3	0	60	70	60	10
4	0	70	80	70	10
5	0	70	70	70	0
6	0	80	80	80	0

While D affects Y , it does not affect $Y_i(1)$ or $Y_i(0)$!

Illustrative Example: Randomized Experiment

Suppose the subjects were randomly assigned to $D = 0$ and $D = 1$ (also imagine these six subjects actually represent 600 subjects).

i	D_i	Y_i	$Y_i(1)$	$Y_i(0)$	τ_i
1	1	70	70	70	0
2	1	80	80	70	10
3	0	60	70	60	10
4	0	70	80	70	10
5	0	70	70	70	0
6	0	80	80	80	0

Randomization implies that $Y_i(1)$ and $Y_i(0)$ are independent of D , which implies that,

$$E[Y_i(1)|D = 1] = E[Y_i(1)|D = 0] = E[Y_i(1)]$$

$$E[Y_i(0)|D = 1] = E[Y_i(0)|D = 0] = E[Y_i(0)]$$

Why Independence?

i	D_i	Y_i	$Y_i(1)$	$Y_i(0)$	τ_i
1	1	70	70	70	0
2	1	80	80	70	10
3	0	60	70	60	10
4	0	70	80	70	10
5	0	70	70	70	0
6	0	80	80	80	0

With randomized treatment assignment,

- ▶ the $D = 1$ rows are a random sample of the rows, therefore $E[Y_i(1)|D = 1] = E[Y_i(1)] = E[Y_i(1)|D = 0]$
- ▶ the $D = 0$ rows are a random sample of the rows, therefore $E[Y_i(0)|D = 1] = E[Y_i(0)] = E[Y_i(0)|D = 0]$

Warning: Independence only for pre-treatment X

- ▶ If X is affected by D , then the previous results will not hold.
- ▶ Suppose $Y = 1$ indicates a heart attack, $D = 1$ is a blood pressure medicine, and X is post-treatment blood pressure. X will be affected by D and will therefore not be independent of D

$$E[Y_i(1)] = E[Y_i(1)|D = 1] = E[Y_i|D = 1] \text{ and}$$

$$E[Y_i(0)] = E[Y_i(0)|D = 0] = E[Y_i|D = 0]$$

Suppose subjects are randomly assigned to $D = 0$ and $D = 1$ (also imagine these six subjects actually represent 600 subjects).

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4	0	70	80	70	10
5	0	70	70	70	0
6	0	80	80	80	0

What is the average $Y_i(1)$?

$$E[Y_i(1)] = E[Y_i(1)|D = 1] = E[Y_i|D = 1] \text{ and}$$

$$E[Y_i(0)] = E[Y_i(0)|D = 0] = E[Y_i|D = 0]$$

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What is the average $Y_i(1)$?

$$\frac{70 + 80 + 70 + 80 + 70 + 80}{6} = 75$$

$$E[Y_i(1)] = E[Y_i(1)|D = 1] = E[Y_i|D = 1] \text{ and}$$

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What is the average $Y_i(1)$ among the $D = 1$ subjects?

$$E[Y_i(1)] = E[Y_i(1)|D = 1] = E[Y_i|D = 1] \text{ and}$$

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What is the average $Y_i(1)$ among the $D = 1$ subjects?

$$\frac{70 + 80}{2} = 75$$

$$E[Y_i(1)] = E[Y_i(1)|D = 1] = E[Y_i|D = 1] \text{ and}$$

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What is the average $Y_i(0)$?

$$\frac{70 + 70 + 60 + 70 + 70 + 80}{6} = 70$$

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What is the average $Y_i(0)$ among the $D = 0$ subjects?

$$\frac{60 + 70 + 70 + 80}{4} = 70$$

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What is the average Y_i among the $D = 0$ subjects?

$$\frac{60 + 70 + 70 + 80}{4} = 70$$

The ATE is the Difference in Average Potential Outcomes

Suppose subjects are randomly assigned to $D = 0$ and $D = 1$ (also imagine these six subjects actually represent 600 subjects).

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What is the average $\tau_i = Y_i(1) - Y_i(0)$?

The ATE is the Difference in Average Potential Outcomes

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6	0	80	80	80	0

What is the average $\tau_i = Y_i(1) - Y_i(0)$?

$$75 - 70 = \frac{0 + 10 + 10 + 10 + 0 + 0}{6} = 5$$

The Expected SLR slope is the ATE with randomized D

i	D_i	Y_i	$Y_i(1)$	$Y_i(0)$	τ_i
1	1	70	70	70	0
2	1	80	80	70	10
3	0	60	70	60	10
4	0	70	80	70	10
5	0	70	70	70	0
6	0	80	80	80	0

$$\begin{aligned}
 \beta_1 &= E[Y_i | D_i = 1] - E[Y_i | D_i = 0] \\
 &= \frac{70 + 80}{2} - \frac{60 + 70 + 70 + 80}{4} \\
 &= \frac{70 + 80 + 70 + 80}{6} - \frac{70 + 70 + 60 + 70 + 70 + 80}{6} \\
 &= \frac{0 + 10 + 10 + 10 + 0 + 0}{6}
 \end{aligned}$$

How can we connect causal inference to the population/descriptive inference we have been doing?

- ▶ Clarify causal vs descriptive interpretations.
- ▶ Distinguish sample vs population causal inference.
- ▶ Derive the SLR model from the potential outcomes model.

Interpretation of β_1 in a simple regression

Causal:

β_1 is the expected effect of an extra unit of D (e.g., year of education) on Y (e.g., wages).

Descriptive:

β_1 is the expected difference in Y (e.g., wages) between two individuals that have a one unit difference in D (e.g., years of education).

Interpretation of β_1 in a simple regression

Causal:

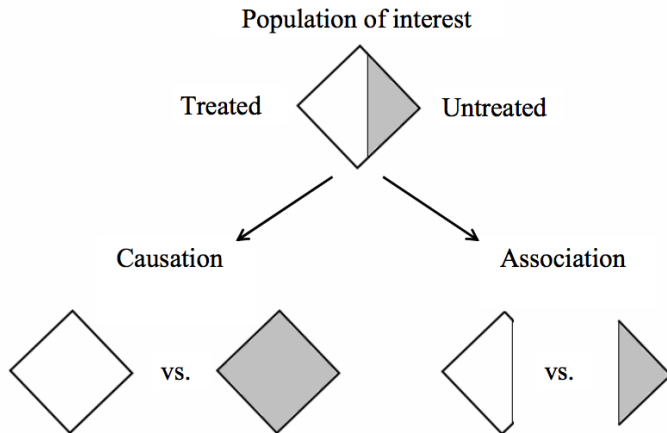
β_1 is the expected effect of an extra unit of D (e.g., year of education) on Y (e.g., wages).

Descriptive:

β_1 is the expected difference in Y (e.g., wages) between two individuals that have a one unit difference in D (e.g., years of education).

If we had a census (the whole population), the descriptive interpretation would always be valid. The causal interpretation would only be valid in some circumstances.

Descriptive (Association) vs Causal from Hernan and Robins



Descriptive vs Causal and Sample vs Pop.

i	D_i	Y_i	$Y_i(1)$	$Y_i(0)$	τ_i
1	1	70	70	?	?
2	1	80	80	?	?
3	0	60	?	60	?
4	0	70	?	70	?
5	0	70	?	70	?
6	0	80	?	80	?
7	?	?	?	?	?
8	?	?	?	?	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Descriptive vs Causal and Sample vs Pop.

i	D_i	Y_i	$Y_i(1)$	$Y_i(0)$	τ_i
Sample Descriptive Statistics			Sample Causal Inference		
1	1	70	70	?	?
2	1	80	80	?	?
3	0	60	?	60	?
4	0	70	?	70	?
5	0	70	?	70	?
6	0	80	?	80	?
Population Descriptive Inference			Population Causal Inference		
7	?	?	?	?	?
8	?	?	?	?	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

What does the ATE tell us?

Let's play CausalDoku!

1. Randomization of D implies (roughly) that the empty $Y_i(1)$ cells for $i = 4, 5, 6$ must be filled with a 4 a 3 and a 2 (in some order)...
2. ...and the empty $Y_i(0)$ cells for $i = 1, 2, 3$ must be filled with a 3 and a 2 and a 1 (in some order).

i	D_i	Y_i	$Y_i(1)$	$Y_i(0)$	$\tau_i = Y_i(1) - Y_i(0)$
1	1	4	4		
2	1	3	3		
3	1	2	2		
4	0	3		3	
5	0	2		2	
6	0	1		1	

CausalDoku

Find two solutions:

1. with constant treatment effects
2. with both negative and positive effects

i	D_i	Y_i	$Y_i(1)$	$Y_i(0)$	$\tau_i = Y_i(1) - Y_i(0)$
1	1	4	4		
2	1	3	3		
3	1	2	2		
4	0	3		3	
5	0	2		2	
6	0	1		1	

CausalDoku

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1	1	4	4		
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3	1	2	2		
4	0	3		3	
5	0	2		2	
6	0	1		1	

Note that both solutions imply the same ATE!